DECISION MEMORANDUM: Strategic PV Integration & US Market Entry URMOSES

To: CEO / CTO

From: Project Lead URMOSES

Date: February 25, 2026

Subject: Approval of PV Sourcing, US-Assembly, and NRTL Certification Strategy

1. EXECUTIVE SUMMARY

To ensure a successful US launch of **URMOSES** (Urban Mobility for Seniors 62+), we propose a hybrid manufacturing model: "**German Engineering – Assembled in USA.**" By utilizing UL-pre-certified high-performance components and a digital assembly workflow, we minimize US import tariffs, reduce OSHA regulatory burdens, and mitigate product liability risks.

2. TECHNICAL SPECIFICATIONS (CTO FOCUS)

The core of the system is designed for maximum energy density and resilience in complex urban environments.

2.1 Component Selection

- **Solar Panels: Aiko Solar (ABC Technology)** or **Maxeon 7 (IBC)**.
  - **Efficiency:** >24% (Industry-leading).
  - **Shading Performance:** Aiko's ABC technology features optimized sub-string management, allowing the panel to produce energy even when partially shaded by buildings or trees—a critical factor for urban senior mobility.
  - **Durability:** No grid lines on the front (Back-Contact) minimizes micro-crack risks from road vibrations.
- **Charge Controller: Genasun GV-10-Li**.
  - **Form Factor:** Ultra-compact (185g), fanless (passive cooling prevents dust/moisture ingress).
  - **Certification:** Pre-certified to **UL 1741**.
- **Interconnects: Stäubli MC4-Evo 2** connectors with **12 AWG UL 4703 PV-Wire**.
  - **Weight Saving:** ~30% lighter than standard 10 AWG cables while meeting strict US fire safety codes (NEC 2023/2026).

3. OPERATIONS & REGULATORY COMPLIANCE (CEO FOCUS)

We avoid the "Made in USA" legal trap (FTC "all or virtually all" rule) by utilizing the legally compliant claim **"Assembled in USA / Made for USA."**

3.1 OSHA Risk Mitigation (Dry-Bonding Strategy)

Instead of liquid industrial adhesives, we will use **3M VHB Solar Tape**.

- **Regulatory Advantage:** Classified as an "Article" under **OSHA 29 CFR 1910.1200**.
- **Operational Impact:** Eliminates the need for specialized ventilation, respiratory protection programs, and hazardous waste disposal. Reduces OSHA compliance costs by **approx. 70%**.

3.2 US Certification Path (NRTL)

We will pursue a **System Listing** via **Intertek (ETL)** or **TÜV Rheinland (cTUVus)**.

- **Cost Efficiency:** Since all components (Panel, Controller, Cable) are already UL-Listed, the certification scope is reduced to "Integration Review."
- **Budget:** Est. **€15,000** (vs. €45,000 for non-certified components).

4. FINANCIAL PROJECTION (Initial Batch: 100 Units + 100 Spares)

Item	Estimated Cost (Total)	Strategic Rationale
PV Hardware (200 Units)	**€28,000 – €34,000**	High-efficiency ABC/IBC cells including volume discount.
NRTL Certification	**€15,000 – €18,000**	One-time investment for US market legal access.
Logistics (Kit Export)	**€5,000 – €8,000**	Sea freight for 200 pre-assembled "Roof Kits."
US Tariffs (Est. 15%)	**€4,500 – €6,000**	Based on 2026 "Assembled in USA" import rates.
TOTAL	**€52,500 – €66,000**	**Investment per vehicle: < €660.**

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5. RISK MITIGATION: DIGITAL QUALITY GATE

To shield the company from US product liability, we will implement a **QR-Code-driven Digital Assembly Log**:

- **Evidence-Based Assembly:** Every bonding process in the US is documented via timestamped photos and sensor data (ambient temperature/humidity) uploaded to our German HQ.
- **Insurance Leverage:** This data serves as the primary defense against claims, proving that US assembly followed the certified German SOP exactly.

6. FINAL RECOMMENDATION

This strategy provides the leanest path to US market entry. It secures technological leadership for URMOSES while protecting the company from the volatile US regulatory landscape.

Action Items for Approval:

1. **Procurement:** Release funds for 200 Aiko/Maxeon Solar Units.
2. **Certification:** Initiate NRTL System Listing with Intertek (Kaufbeuren/USA).
3. **Digital SOP:** Authorize development of the QR-Code Assembly App.